CHEM 427/527 · SESSION 10 · TUESDAY, NOVEMBER 3, 2026

Fourier analysis: how long must the spectrometer listen?

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Drafted with assistance by Claude Code. Reviewed and revised by Matt.

LEARNING OBJECTIVES

By the end of today you can

1. Turn a signal recorded in time into a spectrum with numpy.fft, with an axis in Hz and ppm. Check the spectrum against a known property.
2. Predict where a frequency beyond the sampling limit appears in a spectrum, and confirm it on measured data.
3. Find the shortest acquisition time that still separates closely spaced peaks, against their spacing and width.
4. Choose between stopping a signal abruptly and tapering it to zero, from the artifacts each adds and what the taper costs in time.
5. Write a standing instruction for a coding assistant in CLAUDE.local.md, and assess from a second answer whether it was followed.

FIRST, EVERYONE PULLS

Get session 10 with git pull

`git pull`

S10-fourier appears, with

- **data/fid**: the free induction decay, as the spectrometer wrote it
- **data/acqus**: the spectrometer's acquisition parameters
- **data/records.csv**: 30 records; each of you gets one
- **data/README.md**: where the data come from

Session 9 is due today

- Upload it by 1:30 pm.
- Fill in your prompt record, then Restart and Run All before you upload.

Nothing is downloaded today: the FID and its parameters are in the repository.

FROM COUNTING LONGER TO LISTENING LONGER

What does a longer acquisition buy, and what does it not?

Session 8

Counting longer lowered the detection limit.

Today

An NMR method chooses how long to listen after each pulse.

The question

What does each extra second of listening buy in the spectrum?

More signal, sharper lines, more lines, or nothing?

THE DECISION

How long should a ^{1}H method listen?

Who decides

The scientist who sets up a routine ^{1}H method on a shared spectrometer chooses its acquisition time, AQ.

Too short

Ethanol's 7 Hz triplet merges into one broad line, and a multiplet is misread.

Too long

Listening longer adds no new splitting but still costs instrument time, more so in a 2D experiment.

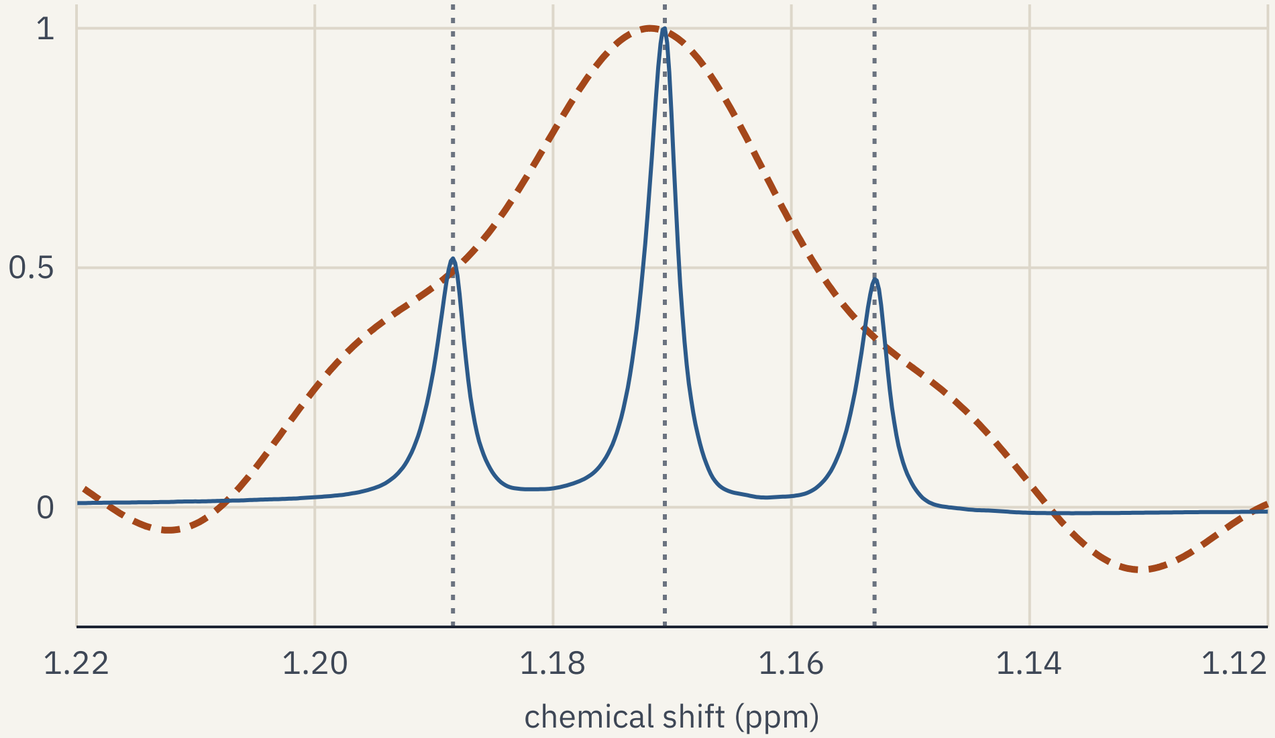
Real: BMRB entry bmse000297

One free induction decay of 100 mM ethanol in D_2O with DSS, one scan at 400 MHz, 3.408 s long. Your record is that FID with seeded noise and extra decay, which mimic a poorer shim.

The spectrometer records a decay in time. The spectrum is its Fourier transform.

LIVE: THE CH_3 TRIPLET, CUT SHORT AND IN FULL

Where does the triplet appear?



First 0.10 s: 481 points, valley 0.98, not resolved

All 3.408 s: 16,384 points, valley 0.07

The three lines: 1.153, 1.171, 1.188 ppm

Each spectrum scaled to its own tallest point. Resolved: the larger valley at most 0.8 of the outer line.

1/AQ is 9.995 Hz at 0.10 s and 0.29 Hz at 3.408 s, against J = 7.08 Hz. 1/J is 0.14 s.

Somewhere between 0.10 s and 3.4 s. Does it depend on the line width?

ASSIGNMENTS

One record each, by student number

#	Record	#	Record	#	Record	#	Record
1	record_01	7	record_07	13	record_13	19	record_19
2	record_02	8	record_08	14	record_14	20	record_20
3	record_03	9	record_09	15	record_15	21	record_21
4	record_04	10	record_10	16	record_16	22	record_22
5	record_05	11	record_11	17	record_17	23	record_23
6	record_06	12	record_12	18	record_18	24	record_24

Your record is your student number from session 4. In Parameters, set **RECORD**, such as RECORD = "record_07", run the Parameters cell, then the cell after TASK 1. Section 1 is the worked example on the real FID. Section 2 cuts your record; section 3 adds a $\cos^2$ taper.

AI PRACTICE 6 OF 8, NEW: KEEP A CLAUDE.MD

Context that the assistant reads every time

Live, on a different question: where is the HDO line?

1. With section 1 run and saved, ask in your own words, naming activity.ipynb in the session folder.

2. Read the answer. What did it not do the way you want every time?

3. Create CLAUDE.local.md in the clone’s top folder, beside CLAUDE.md, and type one line, such as:

Give every chemical shift in ppm to three decimal places, referenced to DSS at 0 ppm, and say which line you used as the reference.

4. Save. Type / and choose Clear conversation, type /memory, send the same prompt, and read the two answers side by side.

A line that works

- Names something an answer visibly does or not: a unit, decimals, a ± and what sets its size, a step to show.
- States a convention for every answer, not a fact about today’s data.
- Can be checked against the next answer, part by part.

“Be precise” does none of the three.

END OF SECTION 1, START OF SECTION 2, THEN TASK 2

Your prompt, your line, your cell

Today you ask for J. Save the notebook before each Ask.

1. End of section 1: add the prompt record, write your prompt on its first line, saying where the spectrum is, and send it.
2. Read the answer against section 1's J: number, unit, decimals, a ± and what sets its size.
3. Start of section 2: write your own line in CLAUDE.local.md, clear the conversation, and send the same prompt.
4. Read the second answer against section 1's J, then fill in the other two lines.

The record, a markdown cell

```
**Prompt record**
- **My prompt:**
- **The line I wrote in CLAUDE.local.md, and what
in the first answer it changes:**
- **What the second answer showed, against section
1's J:**
```

TASK 2: a code cell you write that sets width_b_hz, walking out from the top of your centre line, as section 1 did for Dataset A.

The three lines of your prompt record count toward execution.

THE JUDGMENT CELL, SECTION 4

Your acquisition time, and one entry

(a) Your answer

- What acquisition time would you set for this sample on this spectrometer?
- Your calls under the hard cut and the taper, the negative lobe at the hard cut’s call, and your line width in Hz
- What a wrong call costs either way

(b) One judgment entry

- Your prompt record is the source.
- What you did: changed, rejected, or verified
- The evidence: section 1’s J against the numbers the assistant reported

Which of the two calls would you trust for an assignment, and what would you need to see to decide?

SECTION 1 IS READ AND RUN. SECTION 2 IS YOURS.

Go, then submit

Submit

Fill in your prompt record, then Restart, Run All, save, and upload activity.ipynb by **1:30 pm Thursday, November 5**.

Scoring, 0 to 3

1 if it runs after a restart and holds your prompt record, **0** if not.

0 to 2 for the judgment cell.

Help

Ask Matt, Lucas, or Chris in the room.

The third hour is an office hour.